



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SARADHA RANGANATHAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

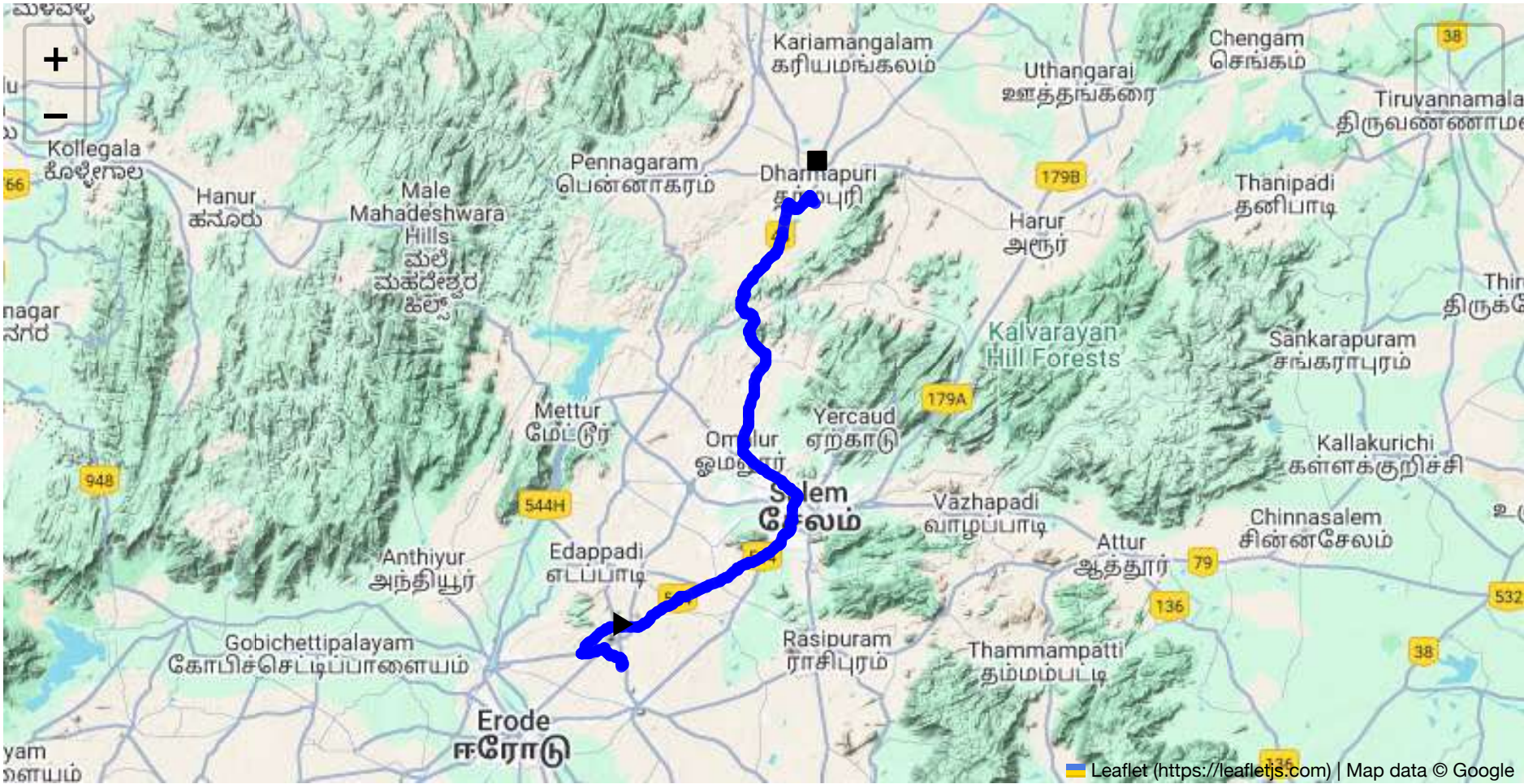


Route Summary:
Total Distance: 116.32 km
Estimated Duration: 2.2 hours

Adjusted Duration (Heavy Vehicle): 2.8 hours

Start: (11.4381, 77.8734)

End: (12.093827, 78.155235)



Welcome to the Journey Risk Management Study

To provide a comprehensive analysis of the route from CVQF+23W, Sangagiri to 35V4+H3V, Dharmapuri via FR48+7XQ, Bhavani Main Road, Padaiveedu, please consider the following points:

- Overview of the Route Map:**
 - The route spans approximately 116.32 kilometers, primarily traversing through Tamil Nadu’s rural and semi-urban areas. It includes key towns and pass-through locations that provide essential amenities but may also present specific challenges typical of Indian regional roads.
- Typical Weather Conditions and Potential Hazards:**
 - Weather:** The region experiences a tropical climate with hot summers, monsoon rains from June to September, and mild winters. Monsoon rains can lead to waterlogged sections, particularly in low-lying areas, which may increase the risk of accidents.
 - Hazards:** Foggy conditions may occur during winter, reducing visibility, and heavy rainfall during monsoon can lead to slippery roads and flooding.
- Traffic Patterns and Congestion:**
 - Peak Hours:** Early morning (7–9 AM) and late afternoon to early evening (5–7 PM) see increased traffic due to local commuting.
 - Congestion-Prone Areas:** The towns of Bhavani and smaller rural intersections may experience congestion, particularly during peak commuting times and market days.
- Road Quality and Infrastructure:**
 - Mixed road quality with some well-maintained stretches and others in need of repair. Urban and suburban sections may have good infrastructure, but rural roads may be narrow and less well-paved, affecting heavy vehicle navigation and increasing journey times.
- Suggestions for Alternative Routes:**

- In case of emergencies or road closures, an alternative route via the NH44 highway can be considered, though this may extend travel time but offers better amenities and infrastructure.

6. Local Regulations Affecting Hazardous Material Transport:

- Stringent regulations require the transport of hazardous materials to comply with safety standards and permit requirements. Local checks at entry and exit points near major towns and intersections.

7. Historical Incidents Involving Heavy Vehicles:

- No major incidents publicly recorded; however, minor accidents involving heavy vehicles and narrow roads have been reported. This suggests a need for caution in navigating narrower sections and sharp turns.

8. Environmental Considerations:

- The route passes through agricultural areas—care is needed to avoid spills that could contaminate local water sources. Also, proximity to small wildlife reserves means drivers should be alert to animal crossings.

9. Communication Coverage:

- Generally good mobile network coverage throughout most of the route but expect dead zones in the more remote and rural sections between towns due to infrastructure limitations.

10. Estimated Emergency Response Times:

- Urban areas can expect quicker response times (30-60 minutes), while rural and remote sections might see delays up to a few hours due to distance and accessibility.

11. Summary of Risk Assessment:

- **High Risk:** during peak hours, monsoon season, in narrow, poorly maintained sections with limited communication.
- **Moderate Risk:** under regular conditions and off-peak times, with potential delays and infrastructure-related challenges.
- Preparedness for adverse weather and clear communication of alternative routes and emergency contacts are crucial in mitigating risks.

This assessment emphasizes the need for careful planning and adherence to safety regulations for the transportation of hazardous materials. The overall caution is recommended due to variable road quality, potential weather impacts, and regulatory constraints.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45853, 77.85139	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	Medium	11.45906, 77.85132	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	High	11.45542, 77.81725	15 KM/Hr
8	Turn	High	11.45636, 77.81678	15 KM/Hr
9	Turn	Medium	11.53573, 77.95794	30 KM/Hr
10	Turn	Medium	11.64183, 78.11810	30 KM/Hr
11	Turn	Medium	11.64338, 78.11998	30 KM/Hr
12	Turn	Medium	11.78039, 78.05792	30 KM/Hr
13	Turn	Medium	11.86753, 78.08331	30 KM/Hr
14	Turn	Medium	11.86981, 78.08387	30 KM/Hr
15	Turn	Medium	11.88041, 78.08143	30 KM/Hr
16	Turn	Medium	11.91302, 78.05856	30 KM/Hr
17	Turn	High	11.95824, 78.04981	15 KM/Hr
18	Turn	High	11.96040, 78.04947	15 KM/Hr
19	Turn	Medium	11.96320, 78.05139	30 KM/Hr
20	Turn	Medium	11.96401, 78.05134	30 KM/Hr
21	Turn	Medium	11.96463, 78.05077	30 KM/Hr
22	Turn	Medium	11.96504, 78.04975	30 KM/Hr
23	Turn	High	11.96597, 78.04920	15 KM/Hr
24	Turn	Medium	11.96679, 78.04942	30 KM/Hr
25	Turn	Medium	11.96804, 78.05080	30 KM/Hr
26	Blind Spot	Blind Spot	12.09202, 78.11459	10 KM/Hr
27	Turn	High	12.08780, 78.12237	15 KM/Hr
28	Turn	High	12.08688, 78.12234	15 KM/Hr
29	Turn	High	12.08464, 78.12765	15 KM/Hr
30	Turn	High	12.10313, 78.14568	15 KM/Hr
31	Turn	Medium	12.10166, 78.14834	30 KM/Hr
32	Turn	High	12.09975, 78.14984	15 KM/Hr
33	Turn	High	12.09643, 78.14897	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
13	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
14	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
18	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
19	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
20	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
21	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
24	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
26	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
27	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
29	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
31	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
32	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
33	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
35	hospital	Sri Sai Sakthi Siddha Hospital	12.0965297, 78.1396218	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
36	clinic	Sri Maruthi Clinic and Scans	12.095525, 78.1386931	30 km/h	Medium
37	clinic	S.K. Clinic	12.0969793, 78.1400279	30 km/h	Medium
38	clinic	Apollo Homeo Clinic	12.104278, 78.147159	30 km/h	Medium
39	clinic	Homoeopathy & Herbal Clinic	12.1040673, 78.1471714	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
8	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
16	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
22	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
23	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
25	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
28	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
30	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
34	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



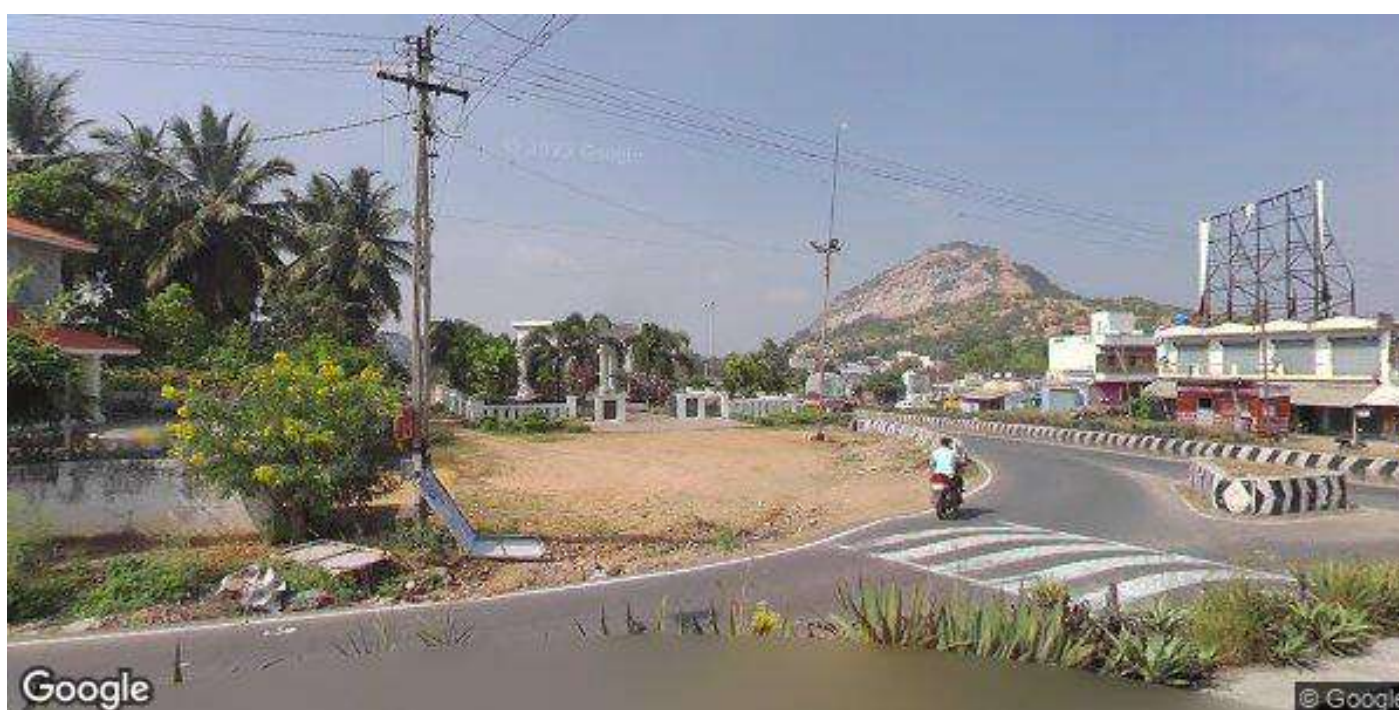
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53573, 77.95794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64338, 78.11998



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78039, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95824, 78.04981



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96463, 78.05077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96679, 78.04942



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.09202, 78.11459



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08780, 78.12237



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08688, 78.12234



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.08464, 78.12765



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.10313, 78.14568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.10166, 78.14834



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.09975, 78.14984



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.09643, 78.14897

